

From On-Shell Amplitudes to Hawking Radiation

Theoretical Developments in Gravitational Wave Physics @ YITP, June 15th, 2026

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Based on arXiv: 2606.13599

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What is a black hole?



- a classical solution of Einstein equation
- macroscopic object
- Hawking radiation (quantum effect)
- information paradox (contradiction to unitarity)

't Hooft suggests a BH model as

Quantum mechanical states whose dynamics are described by unitary S matrix

[G 't Hooft (85,96)]

+

Modern approach of on-shell amplitudes

Outline

- How to describe macroscopic object
- Asymmetry of absorption and emission
- Hawking spectrum

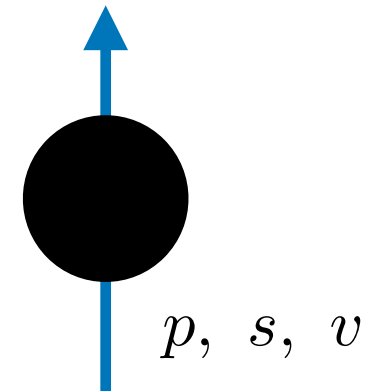
Outline

- How to describe macroscopic object
- Anti-symmetry of absorption and emission
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Set up (microscopic spinning state)

We assume a macroscopic object is represented by one-particle state $|p, s, v\rangle$

- ✓ On-shell $p^2 = M^2$
- ✓ spin (little-group representation) label s
- ✓ Internal label v



Assuming the microcanonical ensemble, microstates are equally probable

$$\hat{1} = \sum_s \sum_v \int d\mu^2 \int d\Phi(p) |p, s, v\rangle \langle p, s, v|$$

$$= \sum_s \int d\mu^2 \int d\Phi(p) \rho(\mu, s) |p, s\rangle \langle p, s|$$

Sum over v
Coarse-graining

$$e^{\mathcal{S}(\mu, s)}$$



Degeneracy of unresolved internal degrees of freedom

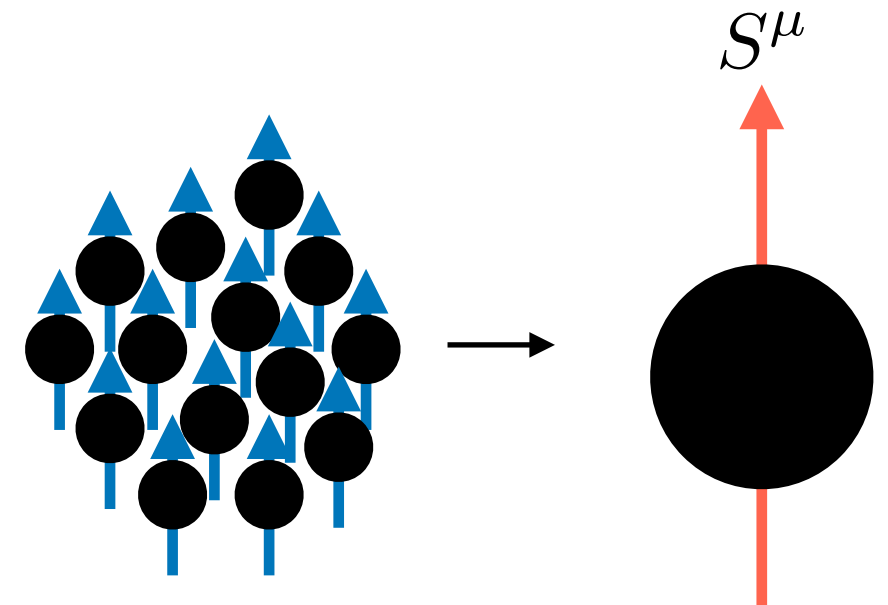
Set up (macroscopic spinning state)

Dynamics of macroscopic spinning dynamics

- ✓ Classical spin vector S^μ
- ✓ Realized by a coherent spin state

$$|p, \alpha\rangle := e^{-\frac{1}{2}\|\alpha\|^2} \sum_{2s=0}^{\infty} \frac{\alpha^{\{I_{2s}\}}}{\sqrt{(2s)!}} |p, \{I_{2s}\}\rangle$$

[R. Aoude and A. Ochirov (08)]



A BH is now described as coherent spin state $|p, \alpha\rangle$ $S^\mu = \mathcal{O}(\hbar\|\alpha\|^2)$

We focus on an object without macroscopic spin ~ slow rotating object

- ✓ $\mathcal{O}(S^0)$ contribution of the small classical spin expansion
- ✓ $|S| \ll GM^2$

Outline

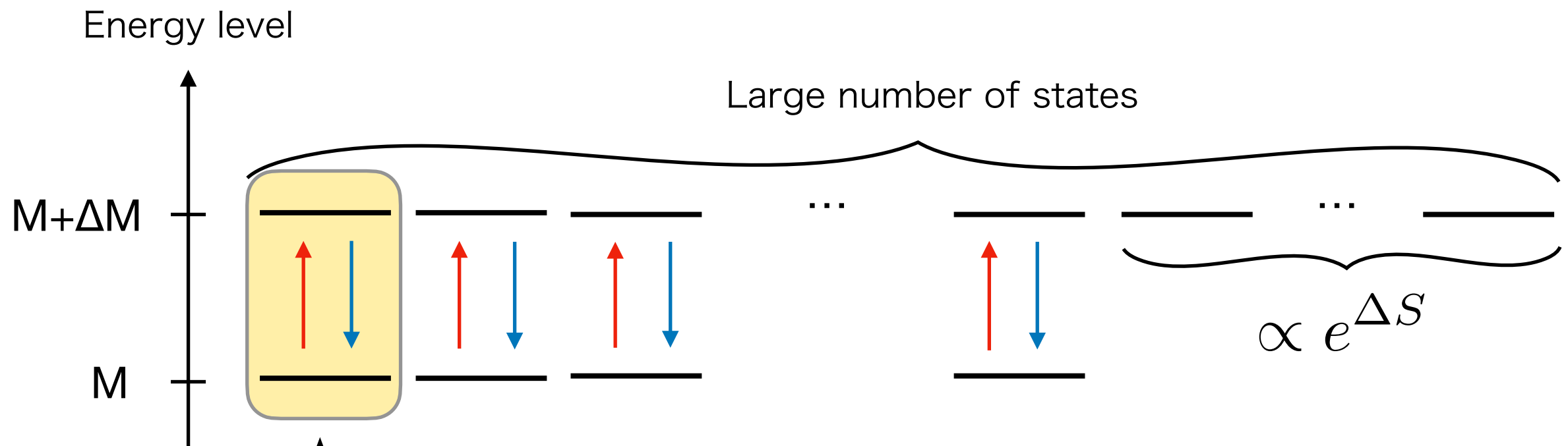
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Micro vs Macro

Let's look at the absorption and emission process



Rough sketch of emission and absorption **without spin**

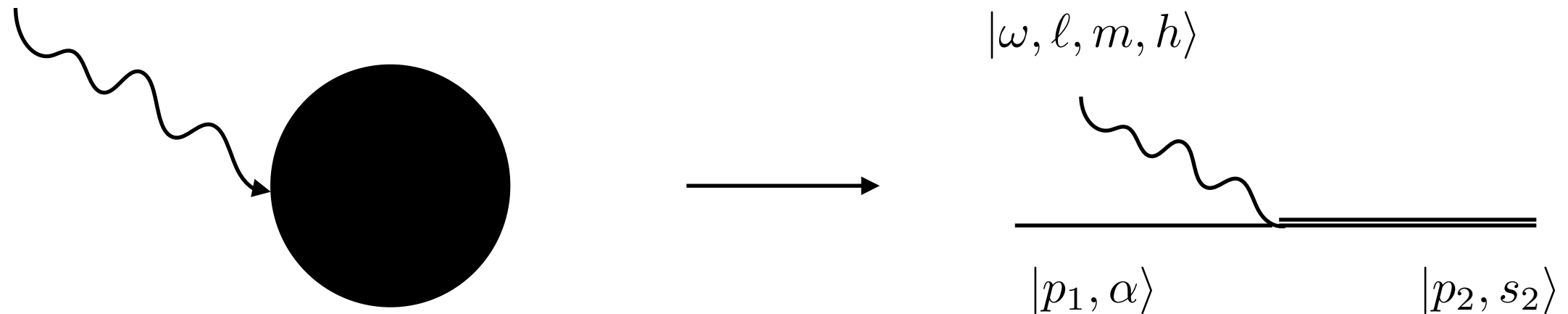


Each **microscopic transition**, abs. & emi.
are related by **time-reversal (CPT)**

Macroscopically,
Two processes are not time-reversal

Macroscopic spinning object

Probability of the absorption process



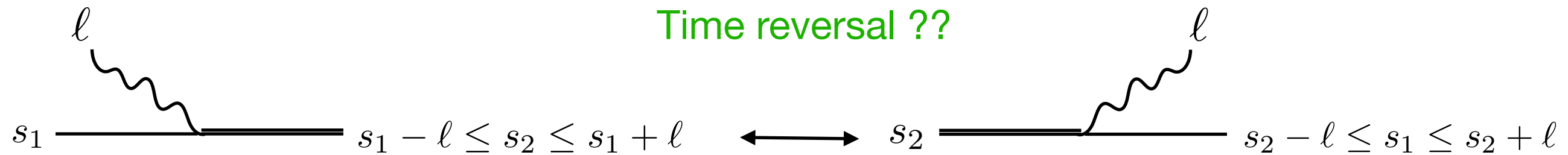
Partial wave probability

$$P_{\ell,m}^{abs} = \frac{|\omega|}{8\pi M_2} \int d^2\hat{\mathbf{k}} d^2\hat{\mathbf{k}}' {}_{-h}Y_{\ell,m}^*(\hat{\mathbf{k}}') {}_{-h}Y_{\ell,m}(\hat{\mathbf{k}}) \frac{\text{Disc}\mathcal{A}_{1',\leftarrow 1}^{-h,h}}{2\pi i}$$

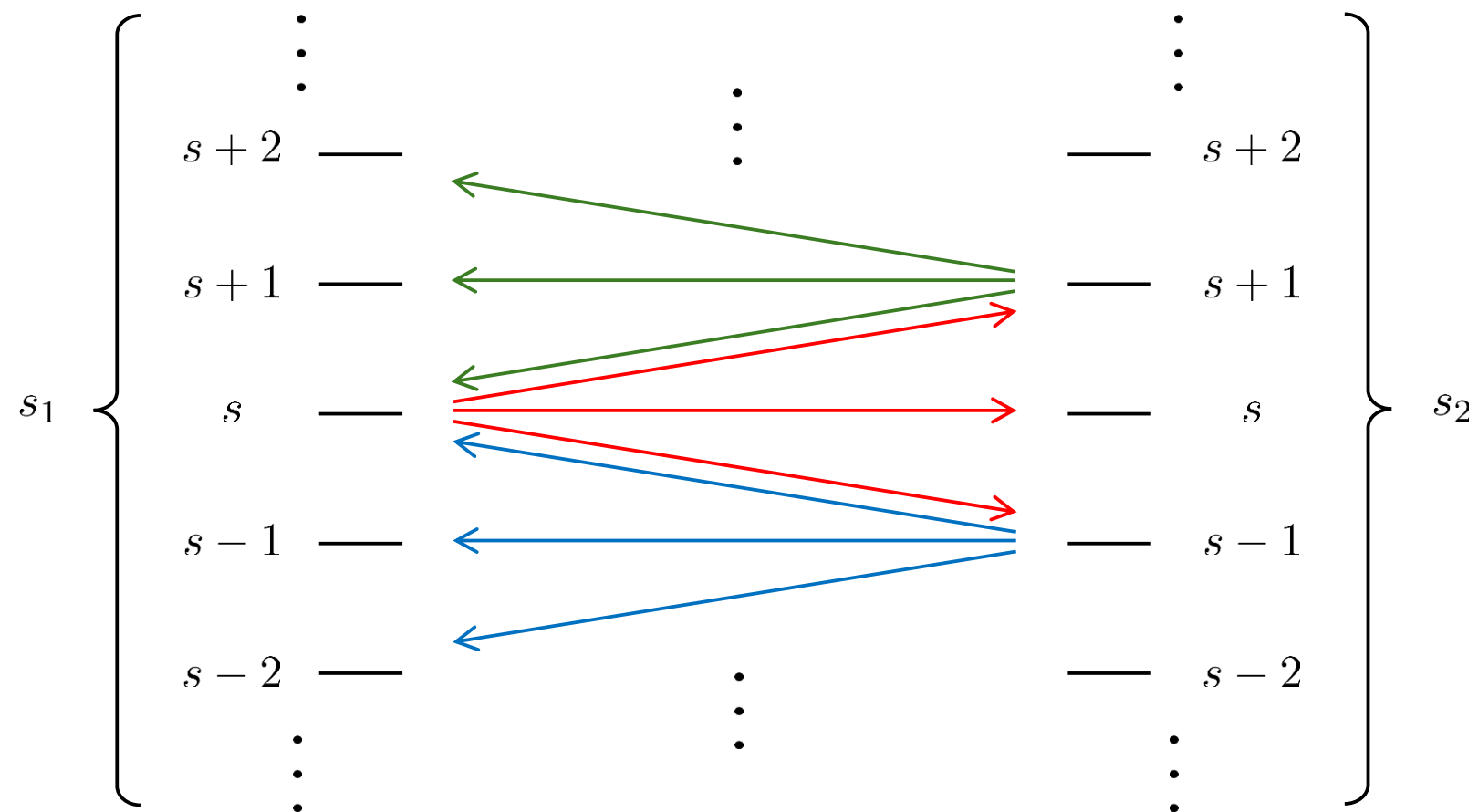
$$\frac{\text{Disc}\mathcal{A}_{1',\leftarrow 1}^{-h,h}}{2\pi i} = e^{-\|\alpha\|^2} \sum_{s_1, s_2, \ell} \frac{\rho(M_2, s_2)}{(2s_1)!} \|\mathcal{A}_{s_1, s_2, h}^{(\ell)}\|^2$$

Subtlety for spinning state

Time reversal ??



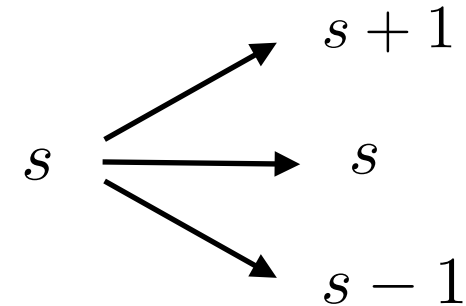
For $l = 1$ case,



Spin Universality

$$\ell = 1$$

$$||\mathcal{A}_s^{(\ell=1)}||^2 = ||\alpha||^{4s-4} \left(|g_{s,s-1}^{(\ell=1)}|^2 \tilde{\mathcal{Y}}^{\tilde{\alpha}\tilde{\alpha}} \mathcal{Y}_{\alpha\alpha} - |g_{s,s}^{(\ell=1)}|^2 \tilde{\mathcal{Y}}^{\tilde{\alpha}}_{\alpha} \mathcal{Y}_{\alpha}^{\tilde{\alpha}} + |g_{s,s+1}^{(\ell=1)}|^2 \tilde{\mathcal{Y}}_{\alpha\alpha} \mathcal{Y}^{\tilde{\alpha}\tilde{\alpha}} \right)$$



By requiring

- **Slow-rotating limit**
- **Macroscopic spherical symmetry** $\rightarrow$ probability $\propto$ Spinning Legendre Polynomial

$$||\mathcal{A}_{s_1,s_2,h}^{(\ell)}||^2 \simeq (-1)^{\ell-\Delta s} |g_{s_1,s_1+\Delta s,h}^{(\ell)}|^2 ||\alpha||^{4s_1-4\ell} {}_{-h}\tilde{\mathcal{Y}}^{\tilde{\alpha}^{\ell-\Delta s}}_{\alpha^{\ell+\Delta s} \ h} \mathcal{Y}_{\alpha^{\ell-\Delta s} \ \tilde{\alpha}^{\ell+\Delta s}}$$

Absorption $|g_{s_1,s_1+\Delta s,h}^{(\ell)}|^2 = |g_h^{(\ell)}|^2 \binom{2\ell}{\ell + |\Delta s|}$

**All $2\ell + 1$ couplings are
Controlled by the single coupling**

Emission $|g_{s_2-\Delta s,s_2,h}^{(\ell)}|^2 = |g_h^{(\ell)}|^2 \binom{2\ell}{\ell + |\Delta s|}$

Spin-universality

Local detailed balance

With spin universality & Slow rotating limit

absorption/emission probability is derived as:

$${}_h P_{\ell,m}^{\text{abs}}(\omega) = \frac{|\omega|}{8\pi M_2} \rho(M_2) |g_h^{(\ell)}|^2 \quad (\omega > 0)$$

$$-{}_h P_{\ell,m}^{\text{em}}(\omega) = \frac{|\omega|}{8\pi M_1} \rho(M_1) |g_h^{(\ell)}|^2 \quad (\omega < 0)$$

$$\frac{{}_h P_{\ell,m}^{\text{abs}}(\omega)}{-{}_h P_{\ell,m}^{\text{em}}(-\omega)} = e^{\Delta\mathcal{S}}, \quad \Delta\mathcal{S} = \mathcal{S}(M_2) - \mathcal{S}(M_1) \quad (\omega > 0)$$

 **Local detailed balance equation**

The ratio of two probabilities are controlled by the entropy difference.

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Application to the BH

BH thermal spectrum is derived by using classical information

(Greybody factor $|\mathcal{T}_\ell|$: classical transmissivity)

When classical wave is sufficiently small & emission is weak

$$|\mathcal{T}_\ell(\omega)|^2 = {}_hP_{\ell,m}^{\text{abs}}(\omega) - {}_{-h}P_{\ell,m}^{\text{em}}(-\omega)$$

Absorption and emission probability of partial wave

$${}_hP_{\ell,m}^{\text{abs}}(\omega) = \frac{|\mathcal{T}_\ell|^2 e^{\omega/T}}{e^{\omega/T} - 1}, \quad {}_{-h}P_{\ell,m}^{\text{em}}(-\omega) = \frac{|\mathcal{T}_\ell|^2}{e^{\omega/T} - 1}, \quad T := \frac{\partial M(S)}{\partial S}$$

Unitarity implies

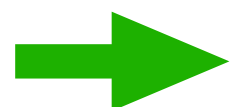
$${}_hP_{\ell,m}^{\text{abs}}(\omega) < 1 \quad |\mathcal{T}_\ell|^2 < 1 - e^{-\omega/T}$$

High energy limit for Schwarzschild

[V. P. Frolov and I. D. Novikov (98)]

$$|\mathcal{T}_\ell|^2 \approx 1 - e^{-8\pi GM\omega}$$

If we assume BH maximize the absorption, we found $T = \frac{1}{8\pi GM}$



We rediscovered the Hawking temperature

Our main result

1. For macroscopic object, we derived the local detailed balance equation with **on-shell approach**.

$$\frac{P^{abs}(\omega)}{-hP^{em}(-\omega)} = e^{\Delta\mathcal{S}}, \quad \Delta\mathcal{S} = \mathcal{S}(M_f) - \mathcal{S}(M_i) \quad (\omega > 0)$$

2. Slow-rotating limit + spherical symmetry → **spin universality**

3. Applying this to black holes and using the **unitarity**, we derived the Hawking thermal spectrum

$$-hP_{\ell,m}^{em}(-\omega) = \frac{|\mathcal{T}_{\ell}|^2}{e^{\omega/T} - 1}$$

$|\mathcal{T}_{\ell}|$: greybody factor

T : Hawking temperature

Summary & Outlook

- We developed non-equilibrium transition between asymptotic equilibrium state.
- Our description is justified only on timescales well below the BH lifetime.
 - Around Page-time, BH can no longer be regarded as an asymptotic states.
 - Resonance?
- It is interesting to consider the connection between our approach and non-equilibrium physics, the eigenstate thermalization hypothesis and fluctuation theorems, and apply them to BH thermodynamics
- It seems possible to construct the framework for studying the two- and many-body dynamics of thermal objects.
 - Radiation from BHs, neutron stars, and their environments can be systematically incorporated into the amplitudes program,
 - It can be opening the possibility of testing these using gravitational-waves observations.